Round Robin - SIVARATHINAM S (241401102)

Activities Terminal Apr 20 17:48

sivarathinam@localhost:~/sivarathinam/scheduling — /usr/bin/vim rr.c

```c
#include <stdio.h>
int main(){
	int n, tq, completed=0, completionTime=0;
	printf("Enter number of processes:\n");
	scanf("%d", &n);
	printf("Enter Time Quantum: \n");
	scanf("%d", &tq);
	int at[n], bt[n], rem[n], tat[n], wt[n];
	float avgtat=0, avgwt=0;
	for(int i=0; i<n; i++){
		printf("Enter AT and BT for P%d: ", i+1);
		scanf("%d %d", &at[i], &bt[i]);
		rem[i]=bt[i];
	}
	while(completed<n){
		int idle=1;
		for(int i=0; i<n; i++){
			if(at[i]<=completionTime && rem[i]>0){
				idle=0;
				if(rem[i]>tq){
					completionTime+=tq;
					rem[i]-=tq;
				} else{
					completionTime+=rem[i];
					tat[i]=completionTime-at[i];
					wt[i]=tat[i]-bt[i];
					rem[i]=0;
					completed++;
				}
			}
		}
		if(idle){
			completionTime++;
		}
	}
	for(int i=0; i<n; i++){
		avgtat+=tat[i];
		avgwt+=wt[i];
	}
	printf("\nAverage Turnaround Time: %.2f\n", avgtat/n);
	printf("Average Waiting Time: %.2f\n", avgwt/n);
	return 0;
}
```

Output :

Activities Terminal Apr 20 17:50

sivarathinam@localhost:~/sivarathinam/scheduling

```
[sivarathinam@localhost scheduling]$ vi rr.c
[sivarathinam@localhost scheduling]$ gcc rr.c -o rr
[sivarathinam@localhost scheduling]$ ./rr
Enter number of processes:
4
Enter Time Quantum:
2
Enter AT and BT for P1: 0 5
Enter AT and BT for P2: 1 4
Enter AT and BT for P3: 2 2
Enter AT and BT for P4: 4 1

Average Turnaround Time: 7.25
Average Waiting Time: 4.25
[sivarathinam@localhost scheduling]$
```